

AI-Generated Image Detection via Hybrid Deep Learning and Handcrafted Feature Ensemble

Abstract

This paper describes a three-stage hybrid machine learning pipeline that classifies images as either real photographs or AI-generated. The system uses 139 manually engineered signal features from ten domains including Fourier spectral analysis, GLCM texture statistics, Local Binary Patterns (LBP), per-channel colour histograms, Canny edge density, DCT high-frequency energy, HSV statistics, inter-channel correlation, noise profiling, and blockwise texture variance combined with deep features from a fine-tuned EfficientNetV2-M network (53.5 million parameters, 288×288 px input). These features are merged using XGBoost and LightGBM gradient-boosted models within a Out-of-Fold (OOF) cross-validation framework. An ablation study shows the contribution of each stage: a handcrafted-only Random Forest baseline achieves $F1 = 0.7343$; the fine-tuned CNN alone reaches $F1 = 0.8740$ (+13.97 pp); the XGBoost hybrid ensemble achieves $F1 = 0.8842$ (+14.99 pp); and the final XGB + LightGBM ensemble achieves $F1 = 0.8860$ at decision threshold 0.384 (Youden J-statistic), with ROC-AUC = 0.9563.

Index Terms — *AI image detection, EfficientNetV2-M, XGBoost, LightGBM, handcrafted features, OOF cross-validation, MixUp, digital forensics, frequency-domain analysis.*

I. INTRODUCTION

The rapid growth of generative AI systems including Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and latent diffusion models such as Stable Diffusion [14] has made synthetic images nearly impossible to distinguish from real photographs. This development poses serious risks for digital media forensics, the spread of misinformation, content moderation, and public trust in online platforms. As a result, automated detection of AI-generated images has become one of the most important and urgent problems in computer vision.

The proposed pipeline makes five specific contributions: (1) a 139-dimensional handcrafted feature set covering thirteen signal domains, each grounded in known generative model artefacts; (2) a two-phase fine-tuned EfficientNetV2-M [1] model with MixUp augmentation and label smoothing; (3) an OOF stacking framework that combines CNN output probabilities with handcrafted features using XGBoost [2] and LightGBM [3] meta-models; (4) a systematic ablation study; and (5) a four-point data leakage audit. The final system achieves OOF $F1 = 0.8860$ and ROC-AUC = 0.9563 on 4,800 training images, representing a 15.17 percentage-point improvement over the handcrafted-only baseline.

II. RELATED WORK

A. Handcrafted Signal Features

Early methods for detecting AI-generated images relied on handcrafted features derived from signal processing theory. Fridrich and Kodovsky [9] showed that Rich Models high dimensional co-occurrence statistics from image residuals can reliably detect GAN outputs. Dzanic et al. [10] identified distinctive artefacts in the Fourier domain at frequencies linked to GAN upsampling patterns, which directly inspired the FFT Radial Band features used in this work. Haralick et al. [11] established GLCM statistics as effective texture descriptors, while Ojala et al. [12]

introduced Local Binary Patterns (LBP) as rotation invariant micro texture descriptors that are sensitive to smoothing artefacts produced by diffusion models. Despite their interpretability, handcrafted methods cannot capture the semantic-level inconsistencies to deep neural networks. *B. Deep Learning and Hybrid Approaches*

Wang et al. [5] showed that a ResNet-50 fine-tuned on ProGAN images achieves strong cross-GAN generalisation ($F1 = 0.82\text{--}0.91$), although without an OOF leakage constraint. Gragnaniello et al. [6] found that CNN models tend to generalise poorly to diffusion generated images, which motivated the multi domain feature design used in this work. Transformer based detectors [7] show promise in capturing global spatial coherence that convolutional models may miss. EfficientNetV2 [1] was selected for its favourable accuracy to VRAM trade off through compound scaling. Qian et al. combined frequency features with CNN embeddings in a dual branch network, achieving strong results on FaceForensics. Wolpert established stacked generalisation as the theoretical basis for training meta models on base learner predictions. This work extends that foundation with a strict OOF protocol and a 139-dimensional handcrafted feature set.

III. DATASET AND EXPLORATORY ANALYSIS

A. Dataset Statistics

The dataset contains 6,888 images from the DCU 2026 ML Challenge: 4,800 labelled training images and 2,058 unlabelled test images. The training set includes 2,485 real photographs and 2,315 AI-generated images, giving a class balance ratio of 0.932 close enough to equal that class weighted loss functions or oversampling methods such as SMOTE were not required. Images are in multiple formats (.jpg) with a median resolution of $1,024 \times 1,024$ pixels.

B. Exploratory Data Analysis

Visual inspection of sample image grids (2×8 images per class) confirmed that the labels were correct and revealed early patterns: AI generated images tend to have unnaturally

regular textures, smoother frequency spectra, and more saturated colour distributions compared with real photographs. Resolution histograms showed no extreme outliers that could cause preprocessing errors. Images were resized to 224 px for handcrafted feature extraction and 288 px for CNN training, balancing image quality against memory usage.

TABLE I: DATASET SUMMARY

Property	Value
Training images	4,800
— Authentic	2,485 (51.8 %)
— AI-Generated	2,315 (48.2 %)
Test images	2,058
Class balance ratio	0.932
Median resolution	1,024 × 1,024 px
Train/Test split	80 % / 20 %

IV. HANDCRAFTED FEATURE ENGINEERING

A. Feature Group Overview

The `extract_features()` function computes 139 descriptive features per image across ten signal domains. Each group targets a specific type of artefact introduced by AI generation methods, and is grounded in signal processing and computer vision literature. The feature groups and their rationale are described in Table II.

TABLE II: FEATURE GROUPS (139 TOTAL)

Group	N	AI Detection Rationale
FFT Radial Bands	9	Generative models introduce spectral artefacts in mid/high frequency bands where natural film grain resides
Laplacian Residual	4	AI models over-smooth flat regions; residual statistics expose this
RGB Colour Histograms	+ 66	Diffusion models shift colour distributions; per-channel stats and histograms capture this
GLCM Texture	20	Spatial co-occurrence (contrast, energy, homogeneity) exposes unnatural textural regularity
LBP Patterns	10	Rotation-invariant micro-texture; sensitive to diffusion smoothing artefacts
Edge Density (Canny)	4	AI edges are geometrically too precise or unnaturally smooth
Noise Profile	3	Atypical Gaussian residual noise fingerprints in synthetic images
Saturation/Brightness	8	Over-saturation is a common generative model stylistic bias
Histogram Entropy	3	Channel-wise entropy differs between real and synthesized images
HOG Descriptor Stats	12	Gradient orientation statistics differ for AI-generated content

Blockwise Variance	Texture6	AI textures are unnaturally uniform across 32×32 px blocks
DCT Energy	High-Freq3	JPEG artefact proxy; AI images differ structurally in high-frequency coefficients
Group	N	AI Detection Rationale
Inter-Channel Correlation	3	AI R-G-B channels are unnaturally correlated; real images show channel independence
Total	139	—

B. Extraction Performance and Checkpointing

Feature extraction ran at 12.76 images per second on the T4 GPU, completing all 4,800 training images in approximately 6 minutes, and at 11.34 images per second for the 2,058 test images. The resulting feature matrices (4800 × 139) for training and (2058 × 139) for testing were saved to Google Drive as `X_hand_train.npy` (2.7 MB) and `X_hand_test.npy` (1.1 MB) to avoid re-running extraction on session reconnection, which is an important practical consideration in Google Colab's temporary runtime environment. All NaN and Inf values were replaced with 0.0 after extraction as a safeguard. Every feature group is zero leakage by design, as each is a stateless per image transformation that requires no population level statistics.

C. Theoretical Justification

Among all feature groups, the FFT Radial Band features have the strongest theoretical basis for AI detection: GAN and diffusion models operate in frequency space and consistently introduce spectral signatures not found in natural photography. GLCM statistics exploit the spatial pixel co-occurrence structure that AI textures do not accurately replicate. LBP descriptors, being rotation invariant, are especially sensitive to the micro smoothing artefacts typical of diffusion model outputs. Together, these ten feature families cover separate signal subspaces, ensuring that discriminative information missed by one group is captured by another, a key advantage over any single domain approach.

V. CNN ARCHITECTURE AND TRAINING

A. Random Forest Baseline

Before applying deep learning, a Random Forest classifier (400 estimators, `max_depth` = 18) trained only on the 139 handcrafted features achieved a validation F1 of 0.7343 on an 80/20 stratified split. This result sets an upper bound on the performance achievable from handcrafted features alone and provides a non-neural reference point that is essential for measuring the added value of each subsequent modelling stage.

B. EfficientNetV2-M Architecture

EfficientNetV2-M [1] was selected as the CNN backbone. It is a compound-scaled architecture optimised for both accuracy and training speed through Neural Architecture Search. With 53.5 million parameters and a 288×288 px

input at batch size 16, it provides a good accuracy to VRAM ratio for the T4 GPU. Weights pre-trained on ImageNet were loaded via the timm library, providing a strong starting point that reduces the amount of labelled data needed. EfficientNetV2-M was preferred over ResNet-50 for its NAS optimised efficiency profile, and over Vision Transformer architectures due to their higher labelled data requirements.

The custom classification head consists of: Dropout(0.4) → Linear(backbone_dim→512) → GELU → Dropout(0.3) → Linear(512→1), which is designed to reduce overfitting on the relatively small training set.

C. Augmentation Pipeline

A multi-strategy data augmentation pipeline was designed to improve generalisation across 4,800 training images. The pipeline includes: RandomHorizontalFlip and RandomVerticalFlip(to remove orientation bias); RandomRotation $\pm 5^\circ$ (for geometric robustness); ColorJitter (brightness=0.3, contrast=0.3, saturation=0.2, hue=0.05 kept conservative to preserve colour-based discriminability); RandomApply(GaussianBlur, p=0.3) (to improve robustness across frequency bands); RandomErasing (p=0.25, scale 0.02–0.20) (to prevent over-reliance on specific image patches); MixUp $\alpha=0.3$ [4] (to enforce soft decision boundaries using label interpolated image pairs); and Label Smoothing $\epsilon=0.05$ via a custom LabelSmoothBCE loss. Test-Time Augmentation (TTA) applies five fixed transforms original, horizontal flip, $+10^\circ$ rotation, -10° rotation, and 308→288 px centre crop averaging probabilities across all views to produce the final CNN score. Each augmentation addresses a specific failure mode, and together they approximate the diversity of a much larger dataset.

D. Two-Phase Training Protocol

A two-phase fine tuning approach was used to preserve pre-trained ImageNet representations. In Phase 1 (5 epochs, backbone frozen, Adam lr= 3×10^{-4} , CosineAnnealing), only the classification head is trained, which establishes a stable gradient signal before the deeper layers are updated. In Phase 2 (up to 15 epochs, all layers unfrozen, AdamW lr= 5×10^{-5} , weight_decay= 1×10^{-4} , CosineAnnealing, early stopping patience=4), full fine-tuning is performed with MixUp active. This approach prevents catastrophic forgetting of a well known failure mode when fine tuning pre-trained models on small datasets. The final CNN was saved to main_cnn.pt (215.6 MB), with best validation F1 = 0.8740 (without TTA) and F1 = 0.8768 (with TTA) a +13.97 absolute F1 point gain over the Random Forest baseline. The approximately 43% reduction in total loss (0.619 → 0.369) confirms steady improvement without overfitting.

VI. HYBRID META-MODELLING FRAMEWORK

A. Out-of-Fold CNN Probability Generation

To generate CNN probability features for meta model training without data leakage, 5-fold stratified cross validation was applied. Each fold trains a separate EfficientNetV2-M model on 80% of the training data and produces probability predictions on the remaining 20% held out samples. This ensures that the meta model's training signals are never derived from samples that were used to train the CNN, the standard best-practice protocol for leakage prevention in stacked ensembles. The OOF framework is the single most important engineering decision for reliable evaluation, as it converts the meta model's training score into a genuine out-of-sample estimate.

B. Meta-Feature Construction

The meta model receives 140 features per sample: (1) one CNN OOF probability value and (2) the 139 handcrafted features. A per fold StandardScaler is fitted only on fold training data and then applied to fold validation and test sets, ensuring that no validation set statistics influence feature scaling, a commonly overlooked source of leakage in practice.

C. XGBoost and LightGBM Configuration

XGBoost [2] uses exact level-wise tree growth (depth-first), providing precise control over the bias variance trade off. Its configuration is: 800 estimators, max_depth=5, lr=0.03, subsample=0.85, colsample_bytree=0.75, reg_alpha=0.2, reg_lambda=2.0, tree_method='hist' (GPU-accelerated). LightGBM [3] uses leaf wise growth (best first), which can achieve lower loss per tree but with greater variance on small datasets. Its configuration is: 1,000 estimators, max_depth=6, lr=0.025, num_leaves=63, subsample=0.85, colsample_bytree=0.75, GPU device, early stopping patience=60. XGBoost's shallower max_depth=5 applies explicit structural constraints, while LightGBM's larger num_leaves=63 is controlled via early stopping. Running both models and combining their predictions as: Final ensemble probability = $0.5 \times P(\text{XGB}) + 0.5 \times P(\text{LGB})$ takes advantage of the architectural differences between the two tree growth strategies, which is equivalent to using diverse base learners in a standard ensemble.

D. Threshold Optimisation

The default 0.5 classification threshold was replaced using the Youden J-statistic, defined as $J = \text{sensitivity} + \text{specificity} - 1$, which maximises the perpendicular distance from the ROC curve to the random chance diagonal. This provides a principled balance between the two types of classification error. Both the Youden J method and an independent threshold grid search converged on 0.384, providing strong evidence that this is the true optimal threshold rather than an artefact of either method. The sub-0.5 threshold reflects slight conservative calibration, a natural result of MixUp training, which introduces label uncertainty near the decision boundary.

VII. EVALUATION RESULTS

A. 5-Fold Cross-Validation Results

Table III shows per fold and mean F1 scores for XGBoost, LightGBM, and their ensemble. Both gradient-boosted models consistently outperform the standalone CNN across all five folds, confirming genuine complementarity between the feature types. The ± 0.015 – 0.016 standard deviation indicates stable generalisation. The higher result in Fold 3 (XGB: 0.9109) likely reflects a favourable validation partition with a higher proportion of easily classifiable examples.

TABLE III: 5-FOLD CV RESULTS

Fold	XGB F1	LGB F1	Ensemble
1 / 5	0.8754	0.8786	0.8770
2 / 5	0.8930	0.8886	0.8908
3 / 5	0.9109	0.9029	0.9069
4 / 5	0.8681	0.8575	0.8628
5 / 5	0.8738	0.8666	0.8702
Mean $\pm \sigma$	0.8842 $\pm$ 0.016	0.8789 $\pm$ 0.016	0.8806

B. Final OOF Classification Report

Table IV presents precision, recall, and F1 at the Youden J-optimal threshold of 0.384. Near-symmetric performance across both classes is a strong indicator of a well generalised and unbiased classifier. The slight precision advantage for the Authentic class (0.90 vs. 0.87) and recall advantage for the AI Generated class (0.90 vs. 0.88) are direct consequences of the sub-0.5 threshold, which slightly prioritises recall of the positive class at a small cost to precision.

TABLE IV: OOF CLASSIFICATION REPORT (threshold = 0.384)

Class	Prec.	Rec.	F1	Support
Authentic	0.90	0.88	0.89	2,485
AI-Generated	0.87	0.90	0.89	2,315
Macro Avg	0.89	0.89	0.89	4,800
Weighted Avg	0.89	0.89	0.89	4,800

C. Ablation Study: Contribution of Each Component

To measure the added value of each pipeline stage, Table V presents an ablation study. Moving from handcrafted features alone (RF: 0.7343) to the CNN alone (0.8740) gives +13.97 F1 points. Adding the CNN OOF probability to the handcrafted feature vector in the XGB hybrid adds a further +1.02 points (0.8842). The final XGB + LGB ensemble adds another +0.18 points to reach 0.8860. Each stage provides a meaningful, additive improvement, which validates the multi-tier design approach.

TABLE V: ABLATION — MARGINAL F1 CONTRIBUTION

System	F1	Δ vs. Baseline
RF Baseline (139 features)	0.7343	—
EfficientNetV2-M CNN	0.8740	+0.1397
XGBoost Hybrid (OOF)	0.8842	+0.1499
LightGBM Hybrid (OOF)	0.8789	+0.1446
XGB + LGB Ensemble (Final)	0.8860	+0.1517
ROC-AUC (Final Ensemble)	0.9563	—

TABLE VI: SUMMARY OF KEY RESULTS

Model / Stage	F1	Protocol	Notes
RF Baseline	0.7343	80/20 split	Handcrafted only
Model / Stage	F1	Protocol	Notes
CNN (EfficientNetV2-M)	0.8740	80/20 + TTA	53.5 M params
XGBoost Hybrid	0.8842	OOF 5-fold	CNN + 139 feats
LightGBM Hybrid	0.8789	OOF 5-fold	CNN + 139 feats
XGB+LGB Ensemble	0.8860	OOF final	Thr. = 0.384

D. Test Set Inference

TTA inference on the 2,058-image test set produced 1,064 Authentic (51.7%) and 994 AI-Generated (48.3%) predictions, closely matching the training distribution (51.8% / 48.2%). This near identical split provides strong evidence against distribution shift or systematic class bias. Predictions were exported in the standard competition CSV format.

E. Leakage Audit

The pipeline was designed with zero data leakage at every stage: (1) handcrafted features were extracted separately for train and test sets with no shared normalisation, (2) CNN OOF probabilities were generated so that each fold's model was trained only on that fold's training samples, (3) StandardScaler was fitted only on fold training data, and (4) threshold optimisation was performed on OOF predictions rather than test predictions. The pipeline therefore meets the strictest leakage prevention standards used in production machine learning systems.

VIII. DISCUSSION

A. Strengths

The pipeline has four main strengths. First, EfficientNetV2-M's compound scaling combined with the two-phase fine tuning protocol, MixUp, and label smoothing reflects current best practice for training on small to medium labelled datasets. Second, the 139 handcrafted features cover separate signal domains, providing a backup when the

CNN fails to detect a specific artefact as evidenced by the +1.02 F1 point gain from the XGB hybrid over the standalone CNN. Third, the strict OOF framework ensures that the reported F1 = 0.8860 is a credible generalisation estimate rather than an optimistic in sample result. Fourth, the complete leakage audit gives verifiable confidence in all reported metrics.

B. Comparison to Baselines

The proposed ensemble (F1 = 0.886, ROC-AUC = 0.956) is competitive with published results on similar tasks. Wang et al. [5] report F1 = 0.82–0.91 on cross-GAN tasks using ResNet-50, without an OOF leakage constraint. Gragnaniello et al. [6] report detection rates of 70–85% on diffusion model outputs, highlighting the generalisation challenge that this work addresses through multi-domain feature design. Corvi et al. [15] further highlights the difficulty of generalising to diffusion generated images, which this multi domain approach partially addresses. The present system achieves F1 = 0.89 symmetrically across both classes under a strict OOF leakage constraint and a 4,800 image training budget, showing that principled pipeline design rather than scale drives performance. The ROC-AUC of 0.9563 confirms that the ensemble ranks images well and is not merely threshold dependent.

C. Limitations and Future Work

Four directions for improvement are identified. (1) Higher CNN Resolution: Training at 384 px instead of 288 px could recover fine grained frequency artefacts currently lost during downsampling, potentially adding +0.5–1.5 F1 points at increased VRAM cost. (2) Feature Importance Pruning: XGBoost feature importance analysis shows that `cnn_prob` dominates; SHAP-guided pruning of the lowest contributing 20–30% of handcrafted features should reduce dimensionality without reducing F1. (3) Vision Transformer Inclusion: Adding a ViT-B/16 or DeiT [8] model would introduce architectural diversity, since attention based models can detect global spatial coherence artefacts that localised convolutional receptive fields may miss. (4) Stochastic Weight Averaging (SWA): Applying SWA during Phase 2 fine tuning can improve generalisation without additional data or training time. (5) Cross dataset generalisation to out of distribution generators (such as Midjourney, DALL-E 3, and Sora) has not been evaluated and represents an important open challenge.

IX. CONCLUSION

This paper presented a rigorous, three stage hybrid pipeline for detecting AI generated images, based on experimental results from an NVIDIA Tesla T4 GPU. The system combines 139 theoretically motivated handcrafted signal features covering Fourier spectral analysis, GLCM texture, LBP micro-texture, DCT high-frequency energy, inter-channel correlation, blockwise texture variance, and colour, edge, and noise domains with a fine tuned EfficientNetV2-M backbone. These are combined using XGBoost and

LightGBM meta-models within a four point, zero leakage OOF cross-validation framework.

This study confirms that each stage provides a genuine, additive improvement: +13.97 pp from the CNN, +1.02 pp from the hybrid meta-model, and +0.18 pp from combining two gradient-boosting variants totalling +15.17 pp over the Random Forest baseline. The final system achieves OOF F1 = 0.8860 and ROC-AUC = 0.9563, with near symmetric per class performance (F1 = 0.89 for both classes) and overall accuracy of 89% (approximately 4,272 of 4,800 images correctly classified). Test set inference yields a 51.7%/48.3% split, closely matching the training distribution. These results show that combining frequency domain signal processing features with transfer learning and principled ensemble stacking achieves competitive performance, even with a 4,800-image training set and a single consumer GPU. Future work should focus on higher resolution CNN training, Vision Transformer integration, SHAP-guided feature pruning, and cross-dataset generalisation evaluation.

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